

KRISHNA SATHVIKA GANNI

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SKILLS

- **Programming Languages:** Python, SQL, PL/SQL, R
- **Data Analysis & Manipulation:** Pandas, NumPy, Exploratory Data Analysis (EDA), Feature Engineering, RFM Analysis, Statistical Modeling
- **SQL & Databases:** PostgreSQL, SQLite, Google BigQuery, Oracle (PL/SQL), CTEs, Window Functions (LAG, FIRST_VALUE), FILTER Clauses, PERCENTILE_CONT, UNNEST (Nested/Semi-Structured Data), Query Optimization
- **Data Engineering & Pipelines:** ETL/ELT Pipeline Development, REST API Integration, Data Cleaning & Validation, Data Pipeline Automation, Change Data Capture (CDC), Slowly Changing Dimensions (SCD Type 2), Record Hashing, Data Versioning & Audit Trails, Apache Airflow (Workflow Orchestration, DAGs), Big Data Concepts
- **Machine Learning:** Scikit-learn, Supervised Learning (Regression, Classification), Model Training & Evaluation, K-Means Clustering, Ensemble Methods (Gradient Boosting, LightGBM), Probabilistic Modeling (BG/NBD, Gamma-Gamma)
- **Deep Learning & Computer Vision:** PyTorch, YOLOv8, OpenCV, Object Detection, Image Classification, Model Inference
- **Time Series & Forecasting:** ARIMA, Prophet, LightGBM, statsmodels, Trend Forecasting, Model Benchmarking (MAE, RMSE, MAPE)
- **Statistics & Experimentation:** A/B Testing, Hypothesis Testing (Z-test, Chi-Square, Welch T-test, Mann-Whitney U), Power & Sample-Size Analysis, Bayesian Inference, Monte Carlo Simulation, Confidence/Credible Intervals
- **Generative AI & LLM Engineering:** LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Databases (ChromaDB), Embeddings (FastEmbed), Google Gemini API, Agentic Workflow Orchestration
- **MLOps & Model Lifecycle:** MLflow (Experiment Tracking & Model Registry), Model Versioning, Data Drift Detection (Evidently), Automated Retraining Pipelines, Model Monitoring, Model Deployment
- **Backend & APIs:** FastAPI, REST API Development, Streamlit
- **Data Visualization & BI:** Tableau, Power BI, DAX, Looker Studio, Plotly, Chart.js, Dashboard Development, KPI Design
- **Testing & Code Quality:** pytest, Unit Testing, Automated Data Validation, Test Coverage Enforcement, flake8 (Linting)
- **DevOps & Infrastructure:** Docker, Docker Compose, Git, GitHub Actions (CI/CD)

EXPERIENCE

- ML/AI Research Assistant**, New Jersey Institute of Technology (NJIT), Newark, NJ Oct 2025 - Present
- Improved AI-based storm damage risk prediction models for power grid resilience using **TensorFlow** and **PyTorch**, achieving a **15% increase in model accuracy** through dataset preparation, optimization, and **image augmentation** techniques.
 - Supported **Smart Health AI** research initiatives across breast cancer, Parkinson's, Alzheimer's, glioma, and healthcare cost prediction by conducting literature reviews, preparing datasets, and exploring **explainable AI (XAI)** approaches.
 - Co-authored a peer-reviewed **IEEE Xplore** publication on machine learning for storm-related energy transmission safeguarding, with the associated poster earning **3rd Place, Best Poster** at the 2025 NJIT Future Energy Transmission Conference.

PROJECTS

- Enterprise RAG Knowledge Assistant** 2026
Tech: Python, FastAPI, LangChain, ChromaDB, FastEmbed, Google Gemini, Streamlit, pytest, GitHub Actions
- Developed a full-stack **Retrieval-Augmented Generation (RAG)** application that ingests PDF, TXT, and DOCX documents and answers natural-language questions using **source-attributed responses** grounded in retrieved document context.
 - Built a document ingestion pipeline (**LangChain** loaders, recursive text chunking) and a persistent vector search layer using **FastEmbed** embeddings and **ChromaDB**, exposed through a **FastAPI REST API** with multi-model fallback and a **Streamlit** dashboard.
- AI Data Analyst Agent** 2026
Tech: Python, LangGraph, Pandas, NumPy, Plotly, FastAPI, Streamlit, Google Gemini, pytest, Docker, GitHub Actions
- Built a **LangGraph**-orchestrated analytics agent that ingests CSV/Excel files and answers natural-language business questions using a **4-node stateful workflow** (load, analyze, visualize, explain), deployed live on **Render**.
 - Implemented deterministic query classification and **Pandas/NumPy** analysis functions (ranking, trend aggregation, linear forecasting), keeping numerical computation fully separate from LLM output, validated via **pytest** and **GitHub Actions CI**.
- MLOps Model Lifecycle Platform** 2026
Tech: Python, scikit-learn, MLflow, FastAPI, Evidently, PostgreSQL, Docker, pytest, GitHub Actions
- Built an end-to-end **MLOps pipeline** for customer churn prediction, training a **Random Forest classifier** and tracking experiments, metrics, and model versions with **MLflow**.
 - Implemented automated **data drift detection** using **Evidently**, triggering a retraining and re-registration workflow when data distribution shifts, served via **FastAPI** with **100% test coverage** enforced in **GitHub Actions CI**.
- Computer Vision Quality Inspection System** 2026
Tech: Python, YOLOv8, PyTorch, OpenCV, FastAPI, Streamlit, pytest, Docker, GitHub Actions
- Built an end-to-end computer vision application using **YOLOv8** for image-based object detection, generating annotated output images with **OpenCV**.
 - Implemented a **confidence-based quality-grading system**, validated with a dedicated **pytest** suite, deployed live on **Streamlit Community Cloud**, and containerized with **Docker Compose**.

Clinical Trials Landscape Analyzer <i>Tech: Python, SQL, PostgreSQL, Apache Airflow, Tableau, Docker, pytest, GitHub Actions</i> - Built an end-to-end ETL pipeline extracting 5,000+ clinical trials from the ClinicalTrials.gov API into a normalized PostgreSQL database, with 12 SQL analytics views (window functions, PERCENTILE_CONT) powering a Tableau dashboard. - Extended the pipeline with a Change Data Capture (CDC) and SCD Type 2 layer, using SHA-256 record hashing and field-level diffing over daily snapshots, orchestrated via Apache Airflow to track historical trial changes.	2025
PJM Hourly Energy Demand Forecasting <i>Tech: Python, ARIMA, Prophet, LightGBM, FastAPI, Streamlit, Docker, pytest, GitHub Actions</i> - Built a time-series forecasting pipeline predicting 168-hour-ahead electricity demand, engineering lag and calendar features with a leakage-free time-based train/test split. - Trained and benchmarked ARIMA, Prophet, and LightGBM; the gradient-boosting model achieved a 2.33% MAPE, outperforming ARIMA (5.83%) and Prophet (7.35%), served via FastAPI and a live-deployed Streamlit dashboard.	2025
Customer Lifetime Value (CLV) Prediction <i>Tech: Python, Pandas, Lifetimes (BG/NBD, Gamma-Gamma), Scikit-learn, Docker, pytest, GitHub Actions</i> - Built a data cleaning and RFM (Recency, Frequency, Monetary) feature pipeline from raw transaction records, computing per-customer purchase history summaries. - Fit probabilistic BG/NBD and Gamma-Gamma models to predict future purchase frequency and spend into a 12-month CLV estimate, segmenting customers into value-based groups using rule-based logic and K-Means clustering.	2025
A/B Testing and Experimentation Analysis Framework <i>Tech: Python, SciPy, statsmodels, NumPy, Pandas, Docker, pytest, GitHub Actions</i> - Built a reusable A/B testing framework applied to a real-world mobile game retention experiment (~90,000 players), implementing frequentist hypothesis testing (z-test, chi-square, Welch t-test, Mann-Whitney U) and power/sample-size analysis. - Implemented a Bayesian Beta-Binomial comparison (Monte Carlo posterior sampling) and a dual-evidence decision rule combining frequentist and Bayesian results into a deploy/retain/inconclusive recommendation.	2025
End-to-End Sales Performance Dashboard <i>Tech: Python, Pandas, SQL (SQLite), Power BI, DAX, pytest, GitHub Actions</i> - Built a Python ETL pipeline to clean, transform, and enrich 9,994 sales records, engineering features including profit margin and discount buckets, validated with a 10-check pytest suite. - Wrote SQL queries analyzing revenue, regional performance, and discount impact, and built an interactive Power BI dashboard with custom DAX measures for KPIs and trend analysis.	2025
Web Analytics Funnel Analysis <i>Tech: SQL, BigQuery, Python, Looker Studio, pytest, GitHub Actions</i> - Analyzed 108,285 sessions of real-world GA4 e-commerce event data in BigQuery, writing SQL with CTEs, window functions, and nested UNNEST operations. - Built a 6-step session-based conversion funnel identifying the largest drop-off point, visualized through an interactive Looker Studio dashboard.	2025
Student Face Recognition System, Group Project <i>Tech: Python, Flask, MySQL, Machine Learning</i> - Collaborated in a team to develop a facial recognition system to verify student identity during exams, implementing real-time recognition using machine learning algorithms for secure authentication. - Integrated live photo verification with a MySQL-backed database for backend validation, helping reduce impersonation risk and enhance academic assessment integrity.	Jan 2024 - Apr 2024

EDUCATION

New Jersey Institute of Technology (NJIT), Newark, NJ Master of Science in Computer Science	Jan 2025 - Dec 2026 GPA: 3.94/4.0
Ramachandra College of Engineering, Eluru, India Bachelor of Technology in Computer Science and Engineering	Aug 2020 - May 2024 CGPA: 8.25/10.0

CERTIFICATIONS

Python for Data Science , YBI Foundation	2022
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